

**Project Design Phase**  
**Proposed Solution**

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 11 September 2026                            |
| Team ID       | NA                                           |
| Project Name  | Metro Ticket Generating System in ServiceNow |
| Maximum Marks | 2 Marks                                      |

**Proposed Solution:**

| S.No. | Parameter                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | <b>Problem Statement<br/>(Problem to be solved)</b> | Urban metro rail commuters face severe congestion and 15–20 minute delays at station ticket booths and Token Vending Machines (TVMs) during peak hours. Stations struggle with high operational overhead from physical paper tokens, cash-handling bottlenecks, and currency disputes. Moreover, traditional digital booking portals provide static receipt confirmations that cannot be read by optical station turnstile scanners, forcing users back into physical queues.                                                                                                                                                                                                                                                                                                                                                                                       |
| 2.    | <b>Idea / Solution description</b>                  | An automated, end-to-end digital ticketing solution developed on the ServiceNow platform. Accessible via the mobile and desktop Service Portal (/sp), commuters select their origin and destination from reference lookups linked to a custom master station registry (u_metro_station_details). An onChange Catalog Client Script calculates Single and Return journey fares in real time based on route matrices. Upon submission, an onSubmit script intercepts the checkout to trigger an spModal popup containing a custom widget (metro_qr_widget) that renders a dynamic, scannable 250x250 2D QR barcode for contactless turnstile gate validation and automatic backend request logging (sc_req_item).                                                                                                                                                     |
| 3.    | <b>Novelty / Uniqueness</b>                         | <ul style="list-style-type: none"> <li>• <b>Service Portal Sandbox Bypass:</b> Uses an Angular-driven spModal and custom widget architecture to render dynamic, high-contrast QR tokens natively without client-side DOM manipulation restrictions.</li> <li>• <b>Integrated Route Matrix Logic:</b> Client-side scripts evaluate origin-destination sys_ids to dynamically calculate fares and prevent journey booking errors.</li> <li>• <b>Conditional Payment UX:</b> Catalog UI Policies dynamically reveal and mandate custom payment input fields when alternate options like corporate vouchers or third-party wallets ("Others") are selected.</li> <li>• <b>Enterprise IT Governance for Transit:</b> Adapts ServiceNow's IT Service Management (ITSM) architecture for citizen-facing Automated Fare Collection (AFC) and passenger tracking.</li> </ul> |
| 4.    | <b>Social Impact / Customer Satisfaction</b>        | <ul style="list-style-type: none"> <li>• <b>Zero Queue Waiting Time:</b> Eliminates standing in physical ticket queues, allowing passengers to purchase passes on the go.</li> <li>• <b>Environmental Sustainability:</b> Replaces non-biodegradable plastic and single-use paper tokens with a 100% paperless, digital QR system.</li> <li>• <b>Cashless Economy:</b> Promotes transparent, digital payments (UPI, Cards, Net Banking).</li> <li>• <b>Accessibility:</b> Responsive web portal design accessible from any standard mobile browser without forcing passengers to install dedicated, heavy native apps.</li> </ul>                                                                                                                                                                                                                                   |

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| 5. | <b>Business Model<br/>(Revenue Model)</b> | <ul style="list-style-type: none"> <li>• <b>Operational Cost Reduction:</b> Lowers operational expenses by minimizing station ticket booth staffing and eliminating physical token manufacturing, restocking, and machine maintenance costs.</li> <li>• <b>Commission &amp; Digital Convenience Surcharges:</b> Metro operators can earn nominal convenience fees or revenue-share percentages on integrated cashless transactions.</li> <li>• <b>Advertising &amp; Partnership Space:</b> The Service Portal interface and digital ticket popups can host targeted digital advertisements, local transit notices, and retail promotions.</li> <li>• <b>B2B Enterprise Transit Accounts:</b> Bulk and corporate transit booking modules with automated invoicing for corporate employee commutes.</li> </ul> |
| 6. | <b>Scalability of the<br/>Solution</b>    | <ul style="list-style-type: none"> <li>• <b>Transit Network Expansion:</b> Easily scales to accommodate new metro lines, routes, and station nodes simply by adding records to the u_metro_station_details table without refactoring core platform code.</li> <li>• <b>Omnichannel Integration:</b> Architecture is built to support automated ticket dispatch via WhatsApp Business API, SMS (Twilio), and Apple/Google Wallet integration.</li> <li>• <b>IoT Hardware Interoperability:</b> Uses unique sys_id tokenization in QR codes, enabling future inbound Scripted REST APIs to interface directly with physical optical gate turnstiles for sub-second gate clearance.</li> </ul>                                                                                                                  |